

SkillClaw: Collective Skill Evolution for Multi-User LLM Agents

SkillClaw enables LLM agent skills to continuously evolve by aggregating cross-user interaction trajectories and applying ag

PROBLEM

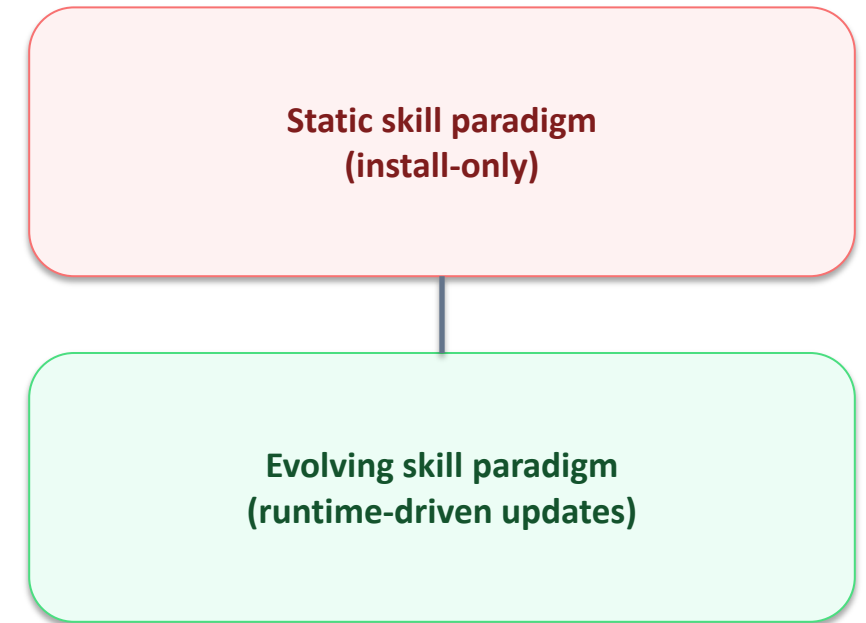
Agent Skills Remain Static After Deployment — Users Re

Reusable skills enable complex tasks, but remain static after deployment.

Users manually install from a centralized hub; fixes discovered during interaction rarely persist.

As tasks recur across users and time, failures/recoveries repeat — no system-level learning.

Memory-based methods are instance-tied; skill libraries often treated as static resources.



ARCHITECTURE

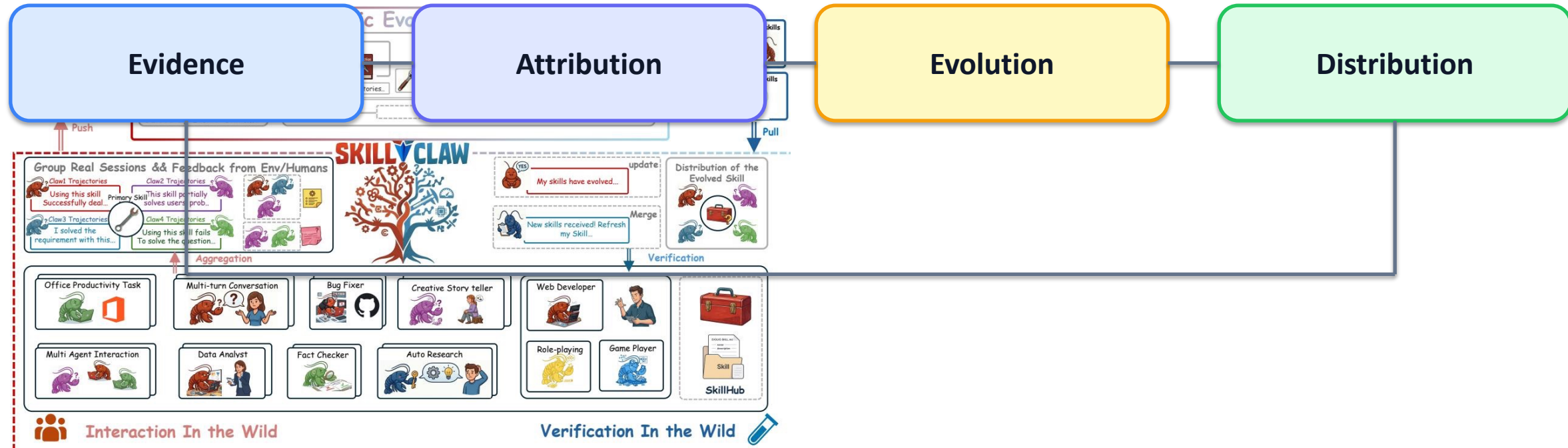
Closed Loop Pipeline: From Multi-User Interaction to Shared Skills

Multi-user interaction → session trajectories

Agents produce structured action–feedback causal chains; grouped by referenced skills.

Verification → shared repository sync

Updated skills are verified before merge and synchronized back to all agents.



Agentic Evolver: Evidence → Attribution → Evolution in T

Evidence: analyze trajectories to identify behavioral patterns.

Attribution: diagnose root causes of failures and successes.

Evolution: propose skill mutations (refine skills or create new ones).

Driven by autonomous, open-ended reasoning over interaction evidence — not predefined update rules.

1) Evidence

2) Attribution

3) Evolution

Output

Verified skills merged and synchronized back to agents.

Cross-User Trajectory Aggregation Exposes Skill Behavior

Trajectories preserve full action–feedback causal chains.

Aggregated across users and grouped by referenced skills → shared evidence base.

Diverse contexts expose skill behavioral boundaries: where it works vs where it breaks.

A single user rarely produces enough signal to separate generalizable improvements from idiosyncratic fixes.

Compatible ecosystems

OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, NemoClaw

Outcome

More robust, evidence-backed updates before distribution.

Case Study: Multimodal Creative Pipeline Skill — 6 Days

Conservative verification

Only 1 of 6 candidates accepted over 6 days (prevents regressions).

Date	Outcome	Rationale (from evolution log)
Day 1	Accepted	Validated /tmp_workspace inputs & environment setup
Day 2	Rejected	Multimodal input validation & output env init
Day 3	Rejected	Multimodal creative pipeline (extract from PDF/video/image)
Day 4	Rejected	Added image classification, visual generation, structured validation
Day 5	Rejected	Creative pipeline + per-file fail-fast validation
Day 6	Rejected	Extended PDF-to-poster / document-to-visual paths

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement

Iterative refinement

3 of 6 days yielded accepted updates before plateauing.

Date	Outcome	Rationale (from evolution log)
Day 1	Accepted	Safe fallback instead of blocking on push failure
Day 2	Accepted	Unified patch/bundle filenames; reduced inconsistency
Day 3	Accepted	Auth-alternative paths & secrets audit; fixed subshell pitfalls
Day 4	Accepted	Same-pool retest; validator confirmed current pool as best
Day 5	Rejected	Added "push hang treated as auth failure"; no improvement
Day 6	Rejected	Added identity config & filename consistency; no improvement

Key Takeaways: Three Properties That Distinguish SkillCl

Collective evolution: individual interactions improve a shared ecosystem.

Fully automatic: runtime interaction drives updates without manual curation.

Agentic evolution paradigm: open-ended reasoning over evidence (not predefined rules).

Implication

Build agent systems that continuously improve as usage accumulates.

Evidence base

Cross-user session trajectories grouped by skill provide the signal for robust updates.